

★ WEEK 7 · PART 3 of 3

Error Correction and Pairs Trading

The Restoring Force, and How to Trade It

The VECM · speed of adjustment · half-life · pairs trading Stage II

Cointegration says the spread reverts; error correction explains WHY. Part 3 turns that restoring force into a working strategy — and confronts the risk that the equilibrium can break.

3

PART 3 · ERROR CORRECTION

The Restoring Force, Traded

Why cointegrated series snap back, the vector error-correction model and its speed of adjustment, the half-life of reversion, and pairs trading built on the spread.

From an Equilibrium to a Mechanism

We proved the tie; now we model the pull

1

We can confirm cointegration

Parts 1 and 2 gave the concept and the tests. We know when a stationary spread — a long-run equilibrium — genuinely exists.

2

But WHAT keeps the spread from wandering?

If the pair deviates, something must pull it back. That something is error correction: an explicit restoring force in the dynamics.

3

And that force is exactly the trade

The speed and reliability of that pull decide whether a spread is tradable. Pairs trading Stage II is built directly on it.

Part 3 in one line: error correction is the restoring force — and the engine of pairs trading.

Error Correction: The Restoring Force

Why cointegrated series snap back

When a cointegrated pair drifts from its equilibrium, the next move tends to close the gap. That built-in correction is what keeps the spread stationary — a self-correcting system.

The mechanism

- A gap opens: the spread moves away from zero.
- The correction responds: subsequent changes shrink the gap.
- The bigger the gap, the stronger the pull back.

Why it must exist

- A stationary spread cannot wander off — so a force must restrain it.
- That force is the error-correction term.
- No such term, no cointegration — the two are inseparable.

Why the Gap Closes

The equilibrium error drives the next move

The feedback

- The system 'remembers' last period's deviation from equilibrium.
- It uses that memory to steer the next change.
- A positive gap prompts a downward correction, and vice versa.

What makes it a system

- Each series may adjust to help close the gap.
- The adjustments are proportional to the size of the deviation.
- Together they hold the pair near its long-run relationship.

The Vector Error-Correction Model

A VAR in differences, plus a memory of equilibrium

$$\Delta \mathbf{Y}_t = \alpha \underbrace{(\boldsymbol{\beta}' \mathbf{Y}_{t-1})}_{z_{t-1} \text{ (last gap)}} + \sum_i \Gamma_i \Delta \mathbf{Y}_{t-i} + \boldsymbol{\varepsilon}_t \quad \text{VECM}$$

The two parts

- The differenced terms are an ordinary VAR on the changes.
- The extra term is the LAST equilibrium error, $z(t-1)$.
- That term is what a plain VAR-in-differences leaves out.

Why it is the right model

- Differencing alone discards the long-run relationship.
- The VECM restores it as an explicit correction term.
- It is the natural model for cointegrated series.

Reading the VECM

One term does all the long-run work

The error-correction term

- It is last period's spread — the deviation from equilibrium.
- Its coefficient decides how strongly the series reacts.
- Everything else is just short-run VAR dynamics.

What the rest handles

- The differenced lags capture short-run momentum and noise.
- They matter for fit but not for the long-run tie.
- The long-run behavior lives entirely in the correction term.

The Speed of Adjustment

How fast the gap closes

$\alpha < 0$: restoring $|\alpha|$ large = fast correction, $|\alpha|$ small = slow

Reading alpha

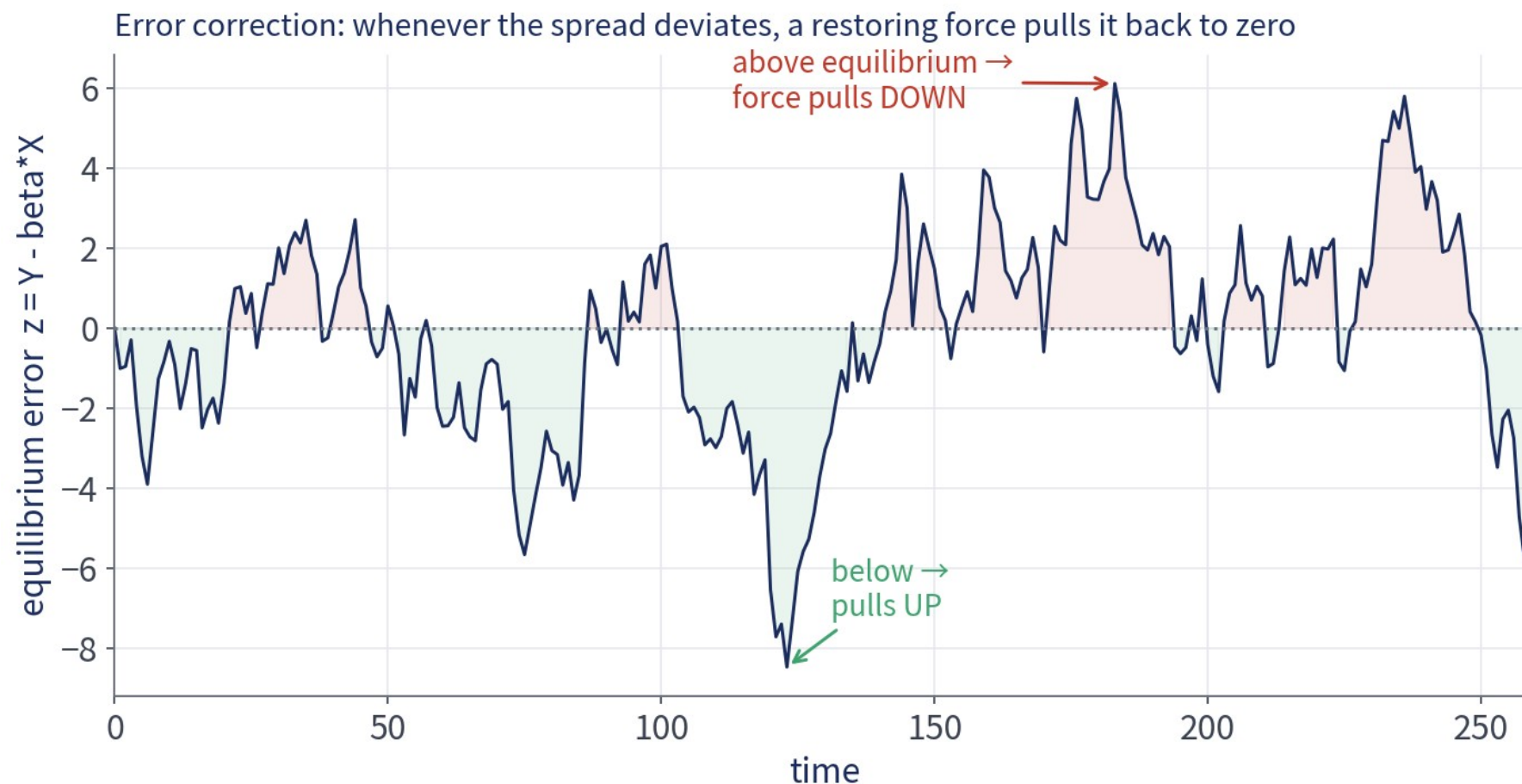
- It multiplies the last equilibrium error in the VECM.
- It must be negative to restore — pushing the gap toward zero.
- Its magnitude sets how much of the gap closes each period.

What it tells you

- A large $|\alpha|$ means fast, reliable reversion — attractive to trade.
- A small $|\alpha|$ means slow reversion — capital tied up longer.
- Which series has the bigger $|\alpha|$ shows who does the adjusting.

The Error-Correction Mechanism

Whenever the equilibrium error strays from zero, the restoring force acts: above the line it is pulled down, below it is pushed up. The gap is never allowed to run away.



Interpreting the Adjustment

Sign, size, and who moves

Sign and size

- Negative alpha restores; positive would be explosive (and rejected).
- Bigger $|\alpha|$ = faster correction, shorter round-trips.
- Near zero = a sluggish, barely-reverting spread.

Who adjusts

- Each series has its own alpha in the system.
- A series with alpha near zero barely moves to close the gap.
- The one with the larger $|\alpha|$ does most of the correcting.

Half-Life of Mean Reversion

How long until half the gap is gone?

$$\text{half-life} = \frac{\ln(0.5)}{\ln(1+\alpha)} \quad (\text{days for half the gap to close})$$

What it is

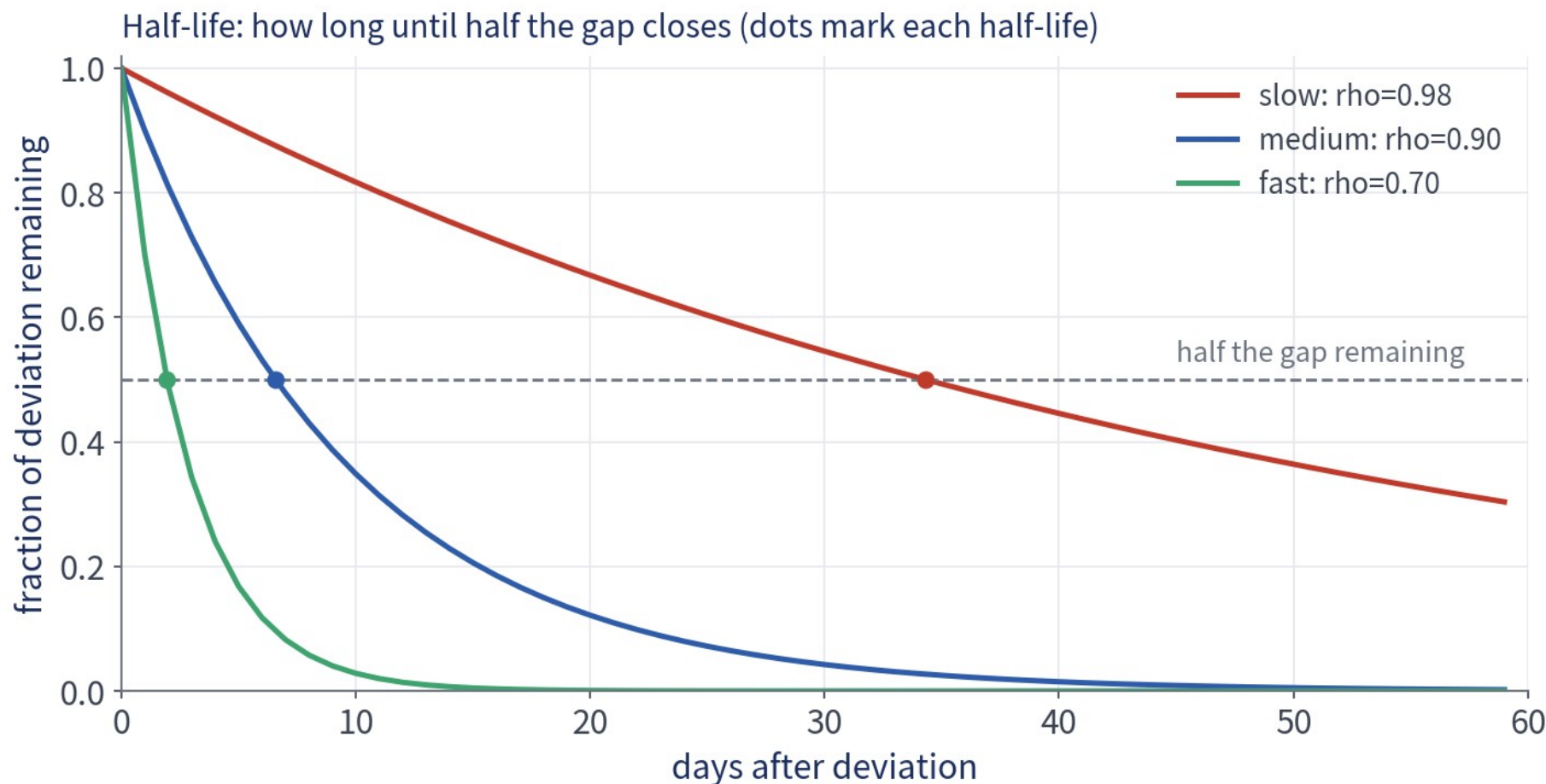
- The time for a deviation to shrink to half its size.
- It converts the abstract alpha into a plain horizon.
- A short half-life means quick, frequent opportunities.

Why traders use it

- It sets a sensible holding period for the trade.
- It flags spreads too slow to be worth the capital.
- It sanity-checks the model: is the speed plausible?

Fast vs Slow Reversion

A deviation decays geometrically; the dots mark each spread's half-life. Faster reversion (green) frees capital quickly; a very slow spread (red) may not be worth trading.



The Granger Representation Theorem

Cointegration and error correction are two sides of one coin

A landmark result: a set of series is cointegrated IF AND ONLY IF it can be written as an error-correction model. The equilibrium and the restoring force always come together.

What it guarantees

- Cointegration $\rightarrow$ an error-correction representation exists.
- An error-correction model $\rightarrow$ the series are cointegrated.
- You never have one without the other.

Why it matters

- It justifies modeling cointegrated series with a VECM.
- It links the long-run equilibrium to short-run dynamics.
- It is the theoretical backbone of this entire week.

Stage II: The Cointegration Spread

The thread gains its long-run anchor

Week 5 gave the pairs thread volatility-scaled bands; now cointegration gives it a genuine anchor. The spread is grounded in an equilibrium, not just a fitted correlation.

What Stage II adds

- A statistically validated, mean-reverting spread.
- A hedge ratio β from the cointegrating relationship.
- A restoring force with a measurable speed and half-life.

The trade in one sentence

- When the spread is stretched, fade it; as it reverts, take profit.
- Long the cheap leg, short the rich leg — market-neutral.
- Exit as the equilibrium error returns to zero.

The Z-Score Signal

Normalize the spread, then act on extremes

$$z_t = \frac{\text{spread}_t - \mu}{\sigma} \Rightarrow \text{enter at } |z_t| \geq 2, \text{ exit near } z_t = 0$$

How it works

- Standardize the spread into a z-score (distance from the mean).
- A large $|z|$ means the pair is unusually stretched.
- Enter against the stretch; exit as z returns to zero.

Why standardize

- It puts every pair on a common, comparable scale.
- Thresholds like 2 sigma become universal entry rules.
- It ties the signal directly to the equilibrium error.

Entry and Exit Rules

Turning the z-score into trades

Entry

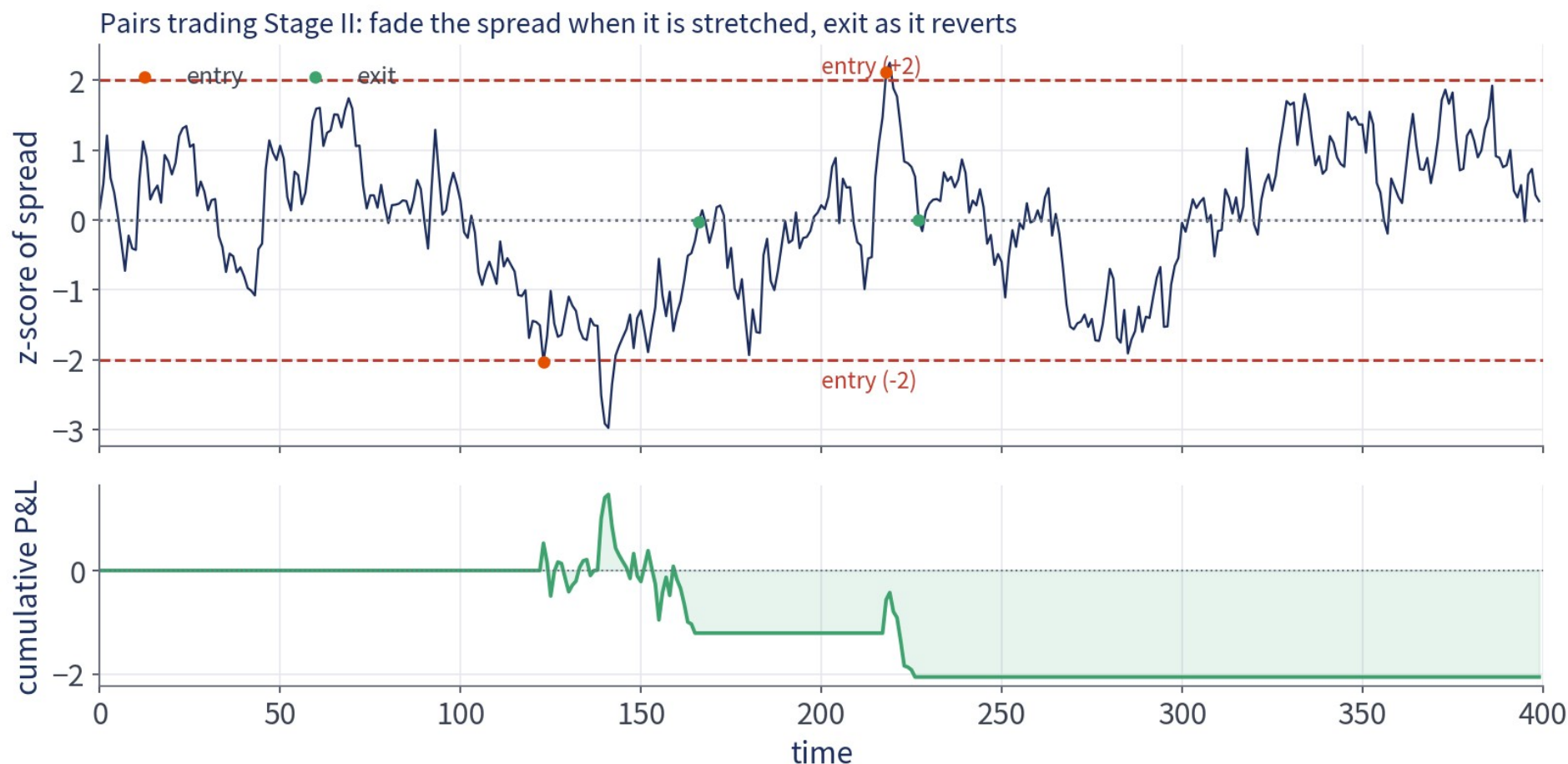
- Enter when the z-score exceeds a threshold (e.g. $|z| \geq 2$).
- Short the rich leg, long the cheap leg.
- Larger thresholds mean fewer but stronger signals.

Exit

- Exit as the spread reverts toward zero (the equilibrium).
- Some exit at $z = 0$; some take partial profits earlier.
- Always pair this with a stop for when reversion fails.

Pairs Trading in Action

Fade the spread when its z-score is stretched past the bands, exit as it reverts to zero. When the equilibrium holds, the cumulative P&L (below) grinds upward.



Sizing and Market Neutrality

The hedge ratio does the balancing

The hedge ratio

- The cointegrating β says how many units of X hedge one of Y.
- Trading in that ratio makes the position market-neutral.
- The bet is on the SPREAD, not on either market's direction.

Sizing the position

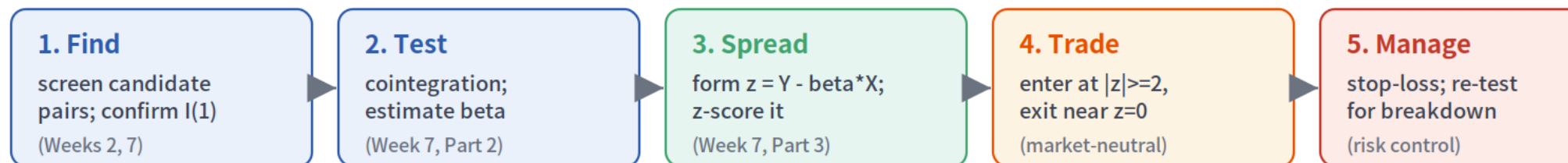
- Scale by the spread's volatility (Week 5) to target a risk level.
- Wider, riskier spreads take smaller notional.
- This is where volatility and cointegration finally combine.

The Complete Pairs-Trading Pipeline

Find a pair, test cointegration, form and z-score the spread, trade the extremes market-neutrally, and manage the risk — with error correction as the engine underneath.

The Complete Pairs-Trading Pipeline

Every week of the course feeds one line - from finding a pair to managing the trade.



The engine underneath

Cointegration gives the anchor (a stationary spread); error correction is WHY it reverts; volatility (Week 5) sizes the bands.

The ever-present risk

Cointegration can break. The spread may stop reverting and drift away - so re-test continuously and always keep a stop.

When Cointegration Breaks Down

The equilibrium is not guaranteed forever

A relationship that held for years can break — a merger, a regulatory change, a shifted business model. When it does, the spread stops reverting and drifts away, turning a 'sure' trade into a loss.

Why it happens

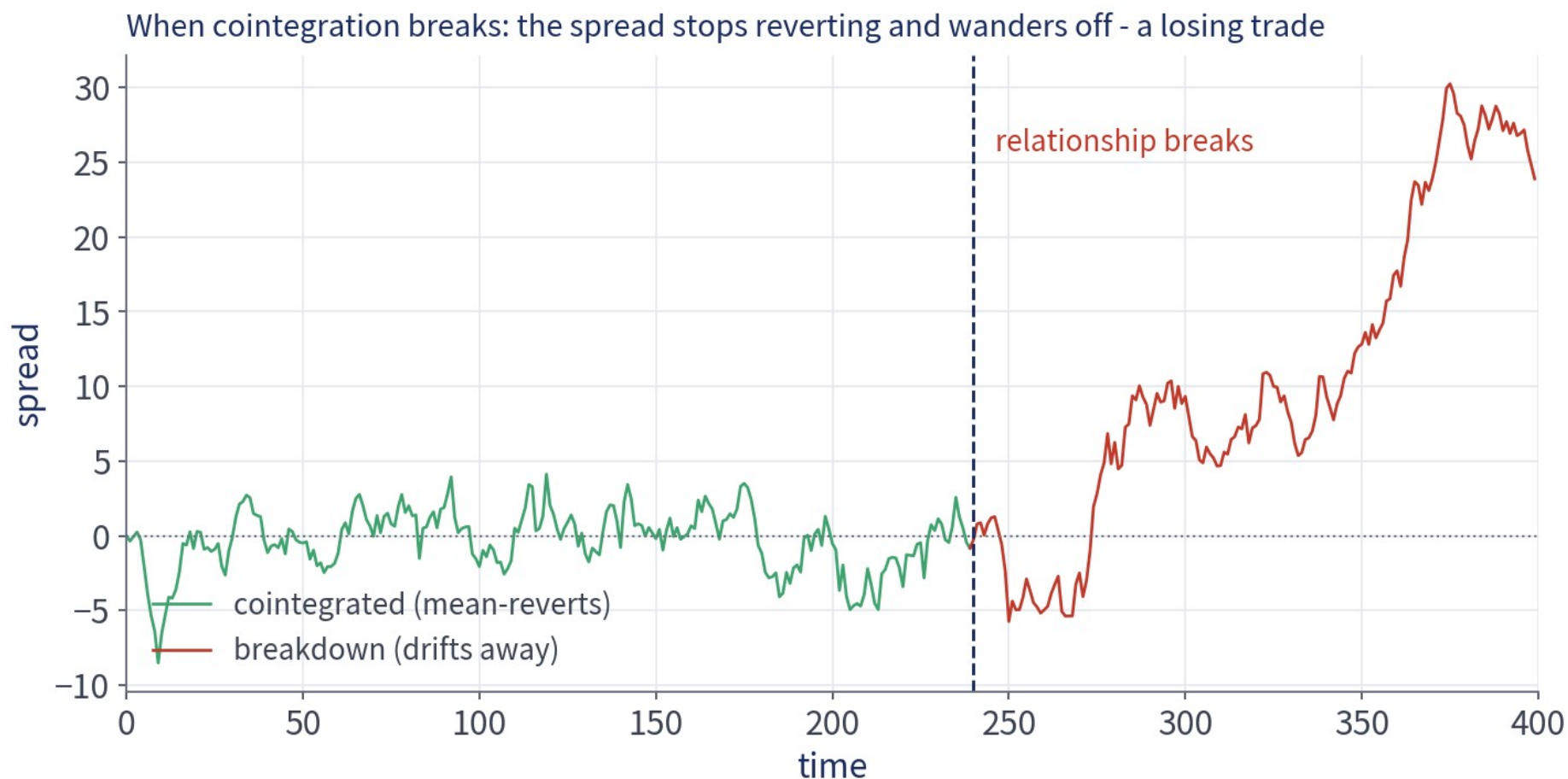
- The economic link that tied the pair changes or disappears.
- Structural breaks, regime shifts, or corporate events.
- The historical β is no longer the right relationship.

The danger

- The spread widens instead of reverting — losses compound.
- 'It always came back before' is a trap.
- This is the defining risk of statistical arbitrage.

When the Equilibrium Breaks

For a while the spread mean-reverts (green). Then the link breaks and it drifts away (red) — a widening, non-reverting spread that no longer offers a trade, only a loss.



Managing the Breakdown Risk

How to trade a spread you don't fully trust

Defenses

- Always run a stop-loss — cap the loss if reversion fails.
- Re-test cointegration regularly; exit if it weakens.
- Diversify across many pairs so one break isn't fatal.

Mindset

- Treat cointegration as a working assumption, not a law.
- Watch for the economic events that could break the link.
- Size positions so a broken pair is survivable.

Build the Spread and Trade It

Direct Claude Code through Pairs Trading Stage II

GOAL From a cointegrated pair, fit a VECM, extract the speed of adjustment and half-life, then back-test a z-score strategy.

[INPUT] Prompt · Claude Code

Write and run `pairs_stage2.py` that:

1. takes a cointegrated pair, forms the spread $z = Y - \beta X$, and fits a VECM;
2. reports the speed of adjustment (α) and the implied half-life of mean reversion;
3. z-scores the spread and back-tests: enter at $|z| \geq 2$, exit near $z = 0$, market-neutral;
4. plots the z-score with signals and the cumulative P&L, and adds a stop-loss.

Report the half-life, the number of trades, and the P&L with vs without the stop.

How to work with it

- α should be negative (restoring).
- A short half-life means more, quicker trades.
- The stop should limit damage if the pair breaks.
- This completes the pairs-trading thread.

Deliverable: `pairs_stage2.py` + a VECM fit and a back-tested spread strategy with a stop.

Three Questions Before You Finish

If you can answer these, Week 7 has landed

Q1 In a VECM, what does a negative alpha coefficient represent?

**Restoring /
adjustment speed**

Q2 A spread's half-life is 3 days vs 60 days. Which is more tradable?

**3 days — faster
reversion**

Q3 Your pair's spread stops reverting and drifts. What has happened?

Cointegration broke

All three come from the speed of adjustment, the half-life, and the breakdown risk.

Three Pitfalls to Avoid

Where pairs trading goes wrong

Common mistakes

- "A cointegrated pair will always revert."
- "No stop-loss needed — the spread must come back."
- "The historical hedge ratio is fixed forever."

The reality

- Cointegration can break; re-test and never assume permanence.
- Always keep a stop — a broken pair drifts without limit.
- The relationship (and β) can drift; monitor and re-estimate.

Week 7 in Three Ideas

Pulling the whole week together

1

Cointegration is a stationary spread of wanderers

Part 1: two $I(1)$ series can share one common trend, leaving a stationary combination — the long-run equilibrium that cures spurious regression.

2

Two tests confirm it; the VECM explains it

Part 2: Engle-Granger and Johansen prove the equilibrium. Part 3: error correction is the restoring force, with a measurable speed and half-life.

3

The spread is the trade — but it can break

Pairs trading Stage II fades the z-scored spread market-neutrally. Its defining risk is that cointegration can break, so always re-test and keep a stop.

Next — Week 8: State-Space Models and the Kalman filter, plus the midterm.